FITNESS TRACKER

Database Management System Project

GUIDED BY: ANJALI SINGHAL MA'AM

PRESENTED BY: ABHASHREE SISOIDA

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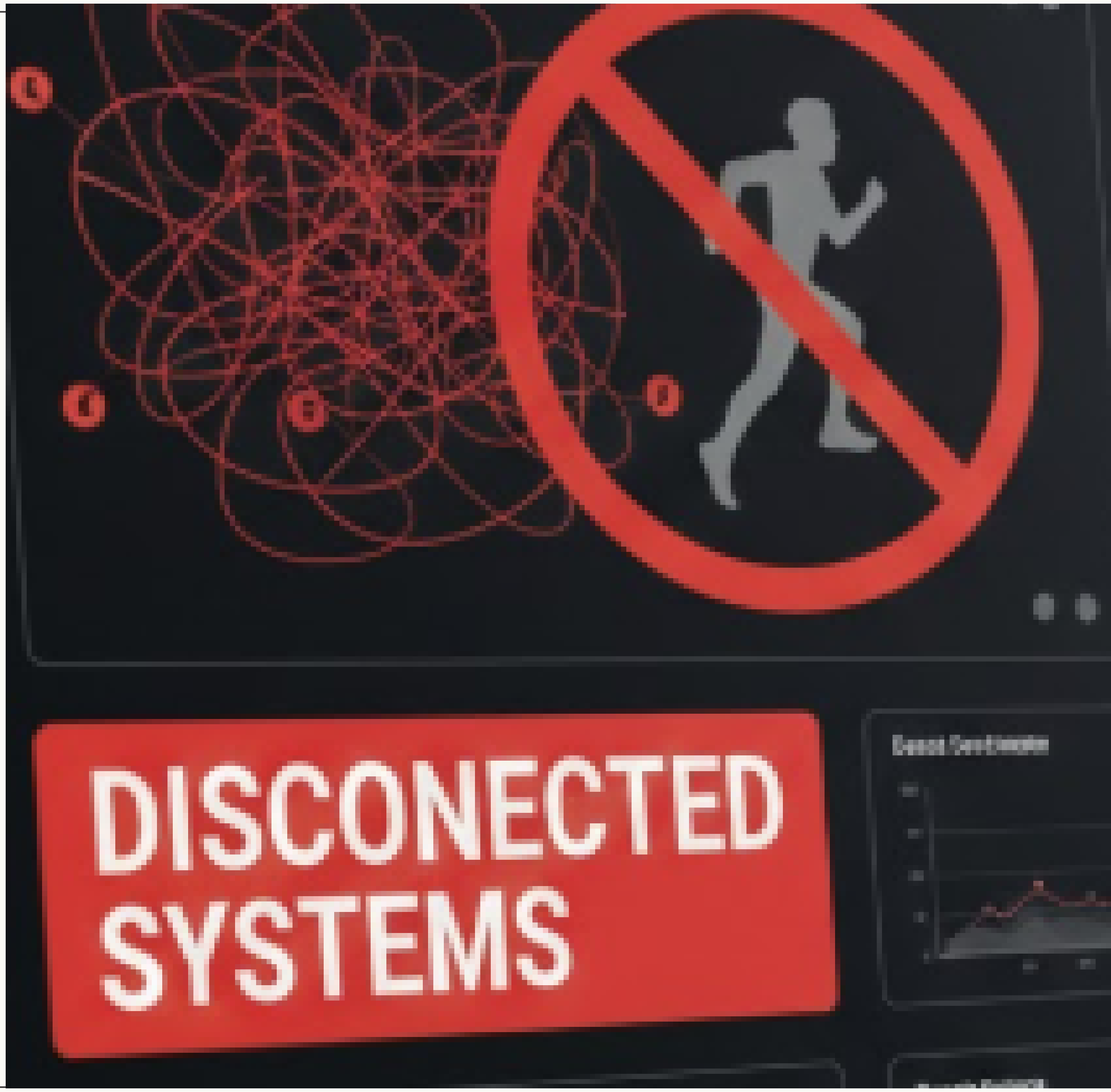
PRESENTED BY: ABHASHREE SISODIA

PROBLEM STATEMENT

Problem
Modern fitness applications collect large volumes of user health data such as activity tracking, workouts, heart rate, sleep patterns, and nutrition records

However, this information is often stored in disconnected systems, making it difficult to maintain structured relationships between data, analyze user performance effectively, and provide timely health insights or automated responses.



OBJECTIVE

THE OBJECTIVE OF THIS PROJECT IS TO DESIGN A CENTRALIZED FITNESS TRACKER DATABASE THAT ORGANIZES FITNESS DATA USING A RELATIONAL STRUCTURE, ENABLING EFFICIENT STORAGE, MEANINGFUL DATA RELATIONSHIPS, AND AUTOMATED MONITORING FOR IMPROVED HEALTH TRACKING AND ANALYSIS.

KEY FEATURES

- USER MANAGEMENT
- DEVICE INTEGRATION
- WORKOUT TRACKING
- HEALTH MONITORING
- SMART AUTOMATION
- EMERGENCY ALERTS
- ACHIEVEMENTS SYSTEM
- DATA ANALYTICS

The Fitness Tracker Database provides a structured platform to manage user fitness activities, wearable device data, workouts, and health records in a centralized system. It enables automated badge rewards, emergency health alerts, and personalized recommendations while supporting analytical insights for monitoring performance and improving overall fitness management.

TECHNOLOGY USED

DATABASE

- MYSQL

TOOLS

- MYSQL WORKBENCH
- CHAT GPT
- CANVA(PRESENTSTION)

CONCEPTS IMPLEMENTED

- Relational Database Design
- Normalization (3NF)
- Primary & Foreign Keys
- Constraints

ADVANCE SQL

- JOIN Operations
- Aggregate Functions
- Triggers
- Stored Procedures

ENTITES

CORE ENTITES

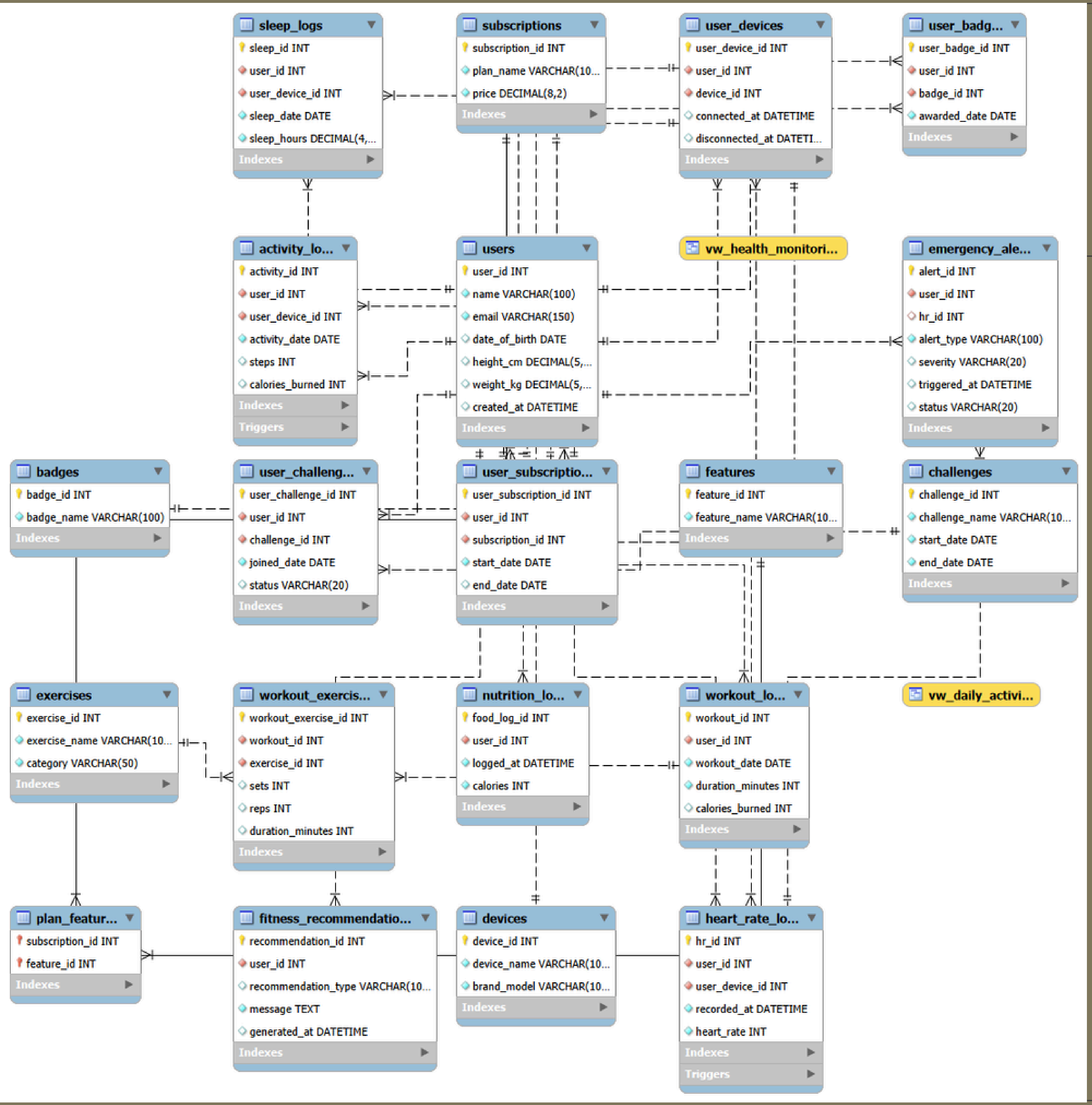
- Users
- Devices
- Exercises
- Subscriptions
- Features
- Badges
- Challenges

RELATIONSHIP ENTITIES

- User Devices
- Workout Logs
- Workout Exercises
- User Subscriptions
- User Badges
- User Challenges

LOG ENTITIES

- Activity Logs
- Heart Rate Logs
- Sleep Logs
- Nutrition Logs
- Emergency Alerts
- Fitness Recommendations



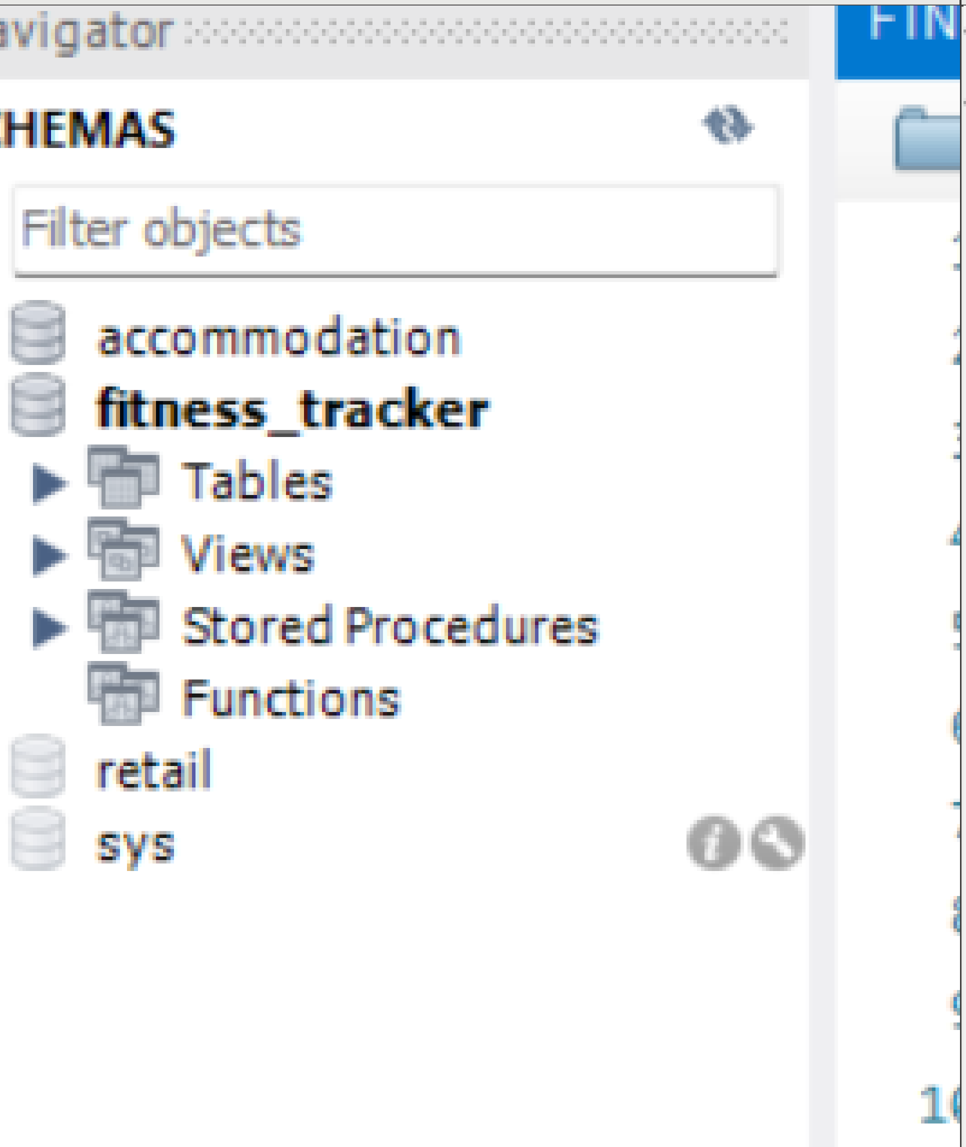
ER DIAGRAM

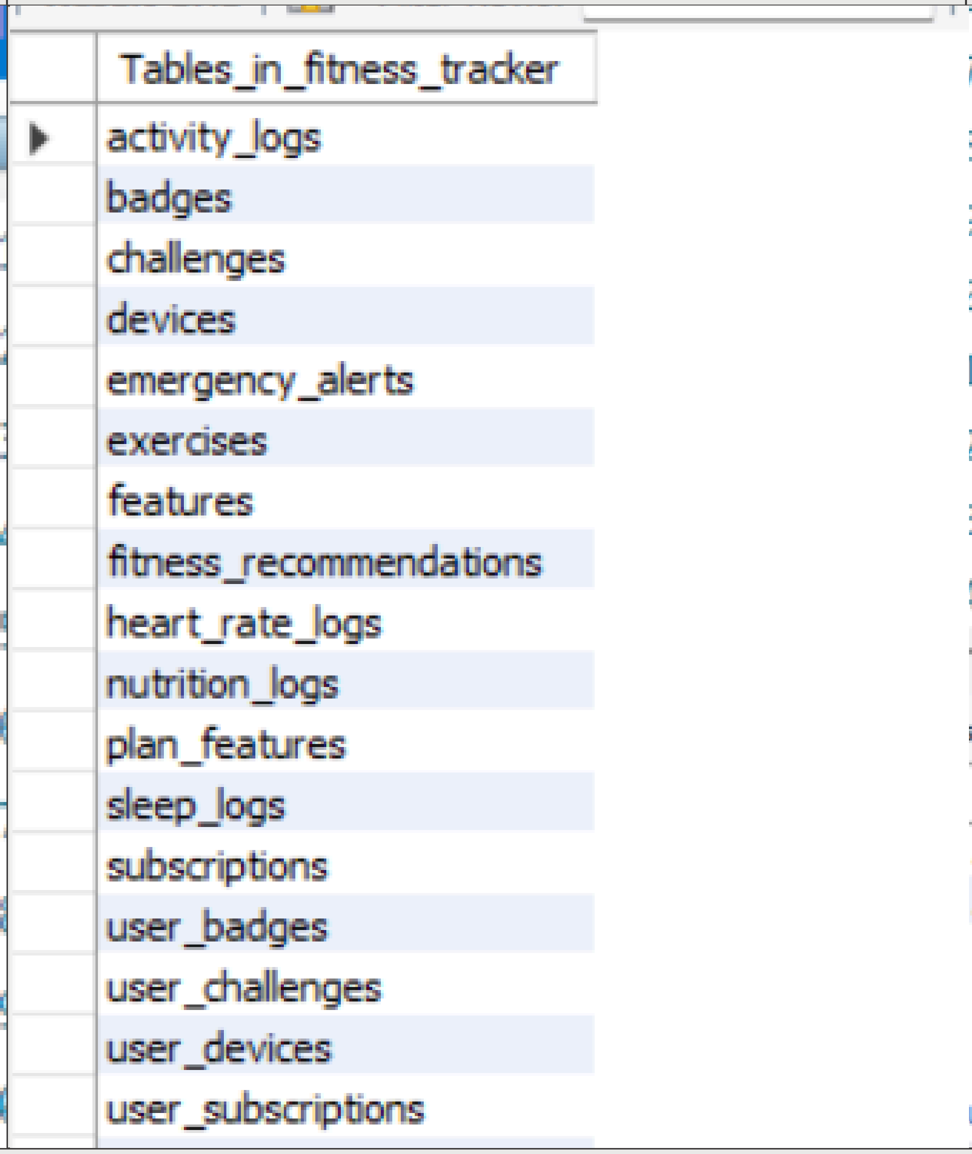
THE ER DIAGRAM REPRESENTS RELATIONSHIPS BETWEEN ENTITIES:

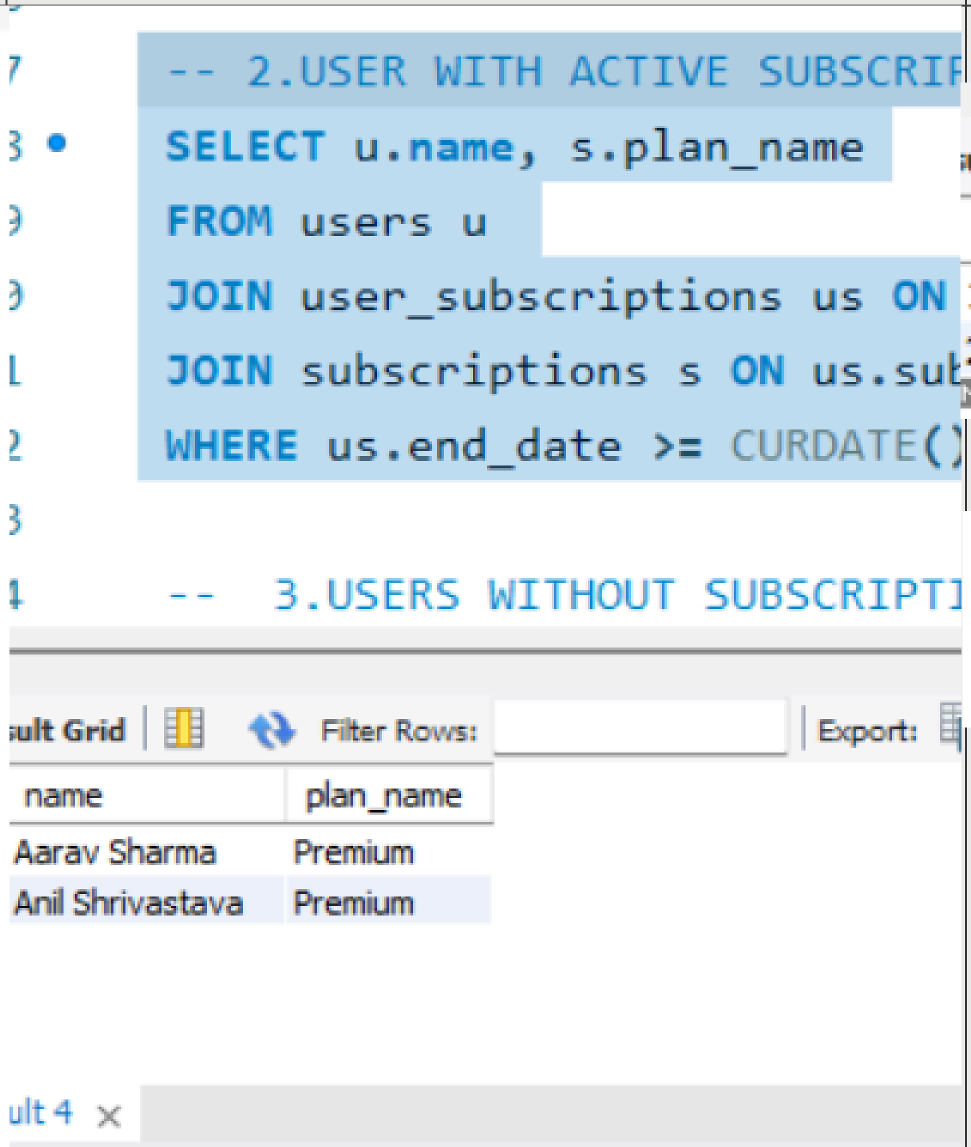
- ONE USER → MANY WORKOUTS
- ONE USER → MANY DEVICES
- ONE SUBSCRIPTION → MANY FEATURES
- WORKOUTS → MULTIPLE EXERCISES
- HEALTH LOGS GENERATED VIA DEVICES

SYSTEM WORKING DEMO

DATABASE CREATED	TABLES GRENERATED	QUERY RUN	TRIGGER

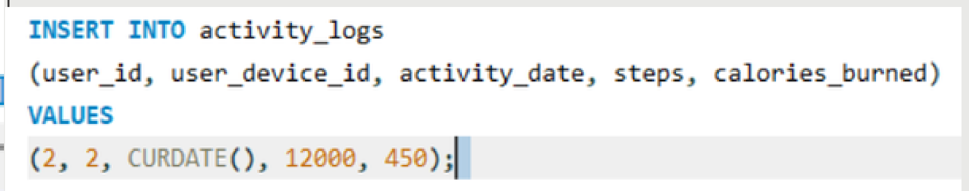




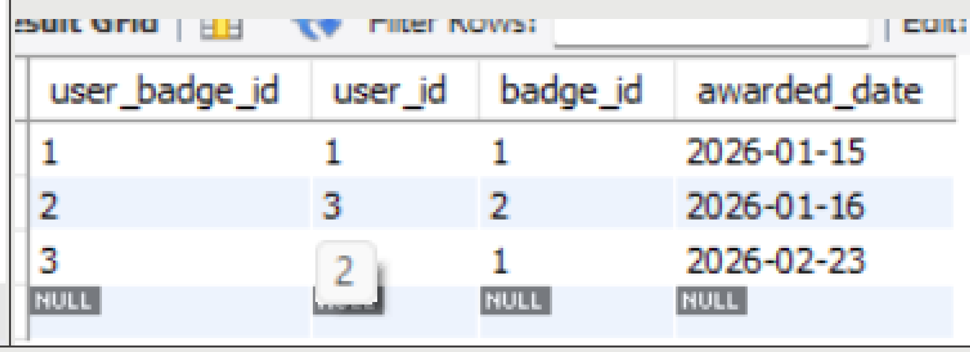


INITIAL BADGE COUNT

USER COMPLETES 10000+ STEPS



BADGE COUNT INCREASES



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FUTURE ENHANCEMENTS

TECHNICAL IMPROVEMENTS
- Mobile App Integration
- Real-time API connectivity
- Cloud database deployment

AI ENHANCEMENTS
- Personalized workout prediction
- Diet recommendation engine
- Health risk prediction

FUTURE EXPANSION
- Social fitness groups
- Trainer dashboard
- GPS activity tracking

CONCLUSION

PROJECT SUMMARY

THIS PROJECT SUCCESSFULLY:

- DESIGNED A NORMALIZED FITNESS DATABASE
- MODELED REAL-WORLD FITNESS TRACKING SYSTEMS
- IMPLEMENTED AUTOMATION USING TRIGGERS
- GENERATED ANALYTICAL HEALTH REPORTS

KEY LEARNING OUTCOMES

- DATABASE DESIGN PRINCIPLES
- RELATIONSHIP MODELING
- SQL OPTIMIZATION
- AUTOMATION USING PROCEDURES

THANK YOU

GUIDED BY: ANJALI SINGHAL MA'AM

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